Egocentric Human Demonstrations for a Bimanual Robot
Part II: Single-Arm Approach to a Cube

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Short version; Part I covers three-cube stacking. Code, results and sample data are available on request.

***Abstract*—We isolate the simplest transferable behaviour, the right arm approaching a red cube, learned from egocentric human demonstrations only. IMU visual–inertial scale is unobservable on these slow motions, so scale comes from a 3.8 cm cube or from the table plane at a taped origin. A shared task frame, a robot wrist hand-eye calibration (0.16 px RMS) and robot-camera rendering put human labels in the robot's frame. Models trained on 80–150 takes overfit from the first checkpoint. Robot runs logged stop reasons but no outcomes, so we report no success rate.**

I. Setup

The operator wears one HandUMI unit [4], a hand-worn UMI [1] variant (wrist fisheye, IMU, jaw encoder) on the right hand; wrist poses come from MASt3R-SLAM [2] (Part I). Labels use Part I's Cartesian contract (16 relative poses, 50.05 ms apart, 15 Hz rows); the left arm and the gripper are dummies masked out of the loss, so 9 of 20 dimensions are supervised. The model is X-VLA (0.9B) [3] with Part I's recipe (soft-prompt slot 20, batch 8, 8-bit AdamW, bf16), trained on an RTX 4090. The robot is the reBot B601 right arm with a wrist C922; the left arm and both jaws are locked during deployment.

II. Metric Scale Without the IMU

HRA_red. 200 takes of "approach the red cube" (10 s) were recorded on 3 October. The per-episode IMU scale has a median of 0.012 and is zero or negative in 95 takes (Fig. 1b): slow approaches do not excite the accelerometer. **Cube PnP.** The static cube (3.8 cm edge) is fitted with two/three-face or near-frontal single-face models and the trajectory scale follows in closed form from the cube staying put. On bimanual HRL80 takes, where the IMU is reliable, 37 of 57 are PnP-valid with a PnP/IMU ratio of 1.062 (p16–p84 0.959–1.274; Fig. 1a). Scale and sanity gates keep 166 of 200 takes (151 train, 20,319 rows; 15 val); median scale 0.354 (p5 0.188, p95 0.500), cube-centre residual 0.17 cm.

III. Putting Human Data in the Robot's Frame

Robot calibration. The head C922 is calibrated on a checkerboard (RMS 1.05 px) and located in the robot base by touching board corners. The robot wrist camera's hand-eye calibration (13 views, 0.16 px RMS) puts it 169.5 mm behind the TCP with its optical axis 33° below TCP x. The IK stack's TCP lies about 82 mm ahead of the physical jaw tip; contact now uses the URDF fingertip box, and earlier "tip below the table" readings were this artefact and were retracted. Linking HRA_red takes to the base after the fact through the head camera reached only 64 of 166 takes, with centimetre-level height spread, and exposed an 18° error in the exported camera–IMU extrinsic.

TABLE I
Held-out loss (masked MSE) by checkpoint; training-subset loss in parentheses.

Data (val takes)	first ckpt	middle	last evaluated
HRA_red v1 (15)	15k: 0.150 (0.066)	30k: 0.198	105k: 0.299 (0.018)
A100, start-anchored (11)	5k: 0.124 (0.110)	30k: 0.141	50k: 0.224 (0.020)
A100, origin-anchored (11)	5k: 0.132 (0.097)	30k: 0.174	45k: 0.190 (0.022)

Robot-camera rendering (v2). v1 rendered wrist views with a guessed camera (62–70° field of view); v2 uses the measured robot camera (52°) and the hand-eye pose, and expresses labels as robot-TCP poses.

Shared origin (HRA_A100). Each take starts with the HandUMI jaw tip held for 2 s on a taped cross that the robot jaw tip also touched, then a 3 s move to a robot-like start and a 10 s approach. Three robot touches on the tape line define the task frame G (collinear within 0.38 mm over 60.4 mm; table at $z = -27.1$ mm). At the origin the table plane in the first MASt3R pointmap gives a second scale source (ratio to cube PnP 0.999, per-take error p50 8%, p90 37%), and the origin yaw is measured per take (error p50 0.21°). 107 takes were kept on 6 October. A table-aware conversion (fingertip floor at table + 5 mm, endpoint lift at most 30 mm) keeps 86 (CT5/CT6), with fingertip clearance p50 12.6 mm, minimum 7.6 mm.

IV. Training

Every run overfits from its first evaluated checkpoint (Table I); the v1 300k run ended at training loss 0.002. Anchoring the state at the take start or at the origin makes no difference to held-out loss. A run on the 85 accepted 6 October takes with the chosen robot start pose (CT7) was in progress when this was written.

V. Robot Runs and Deployment

v1 moved towards the cube from nearby starts and dragged along the table for far cubes (about 39 and 26 cm in two runs).

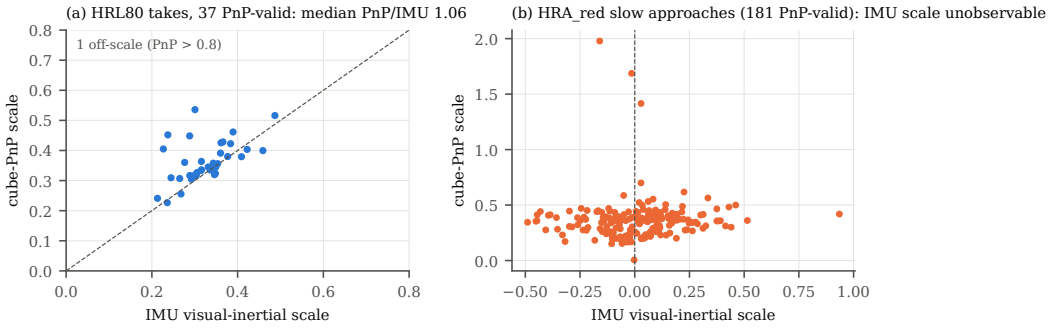

Fig. 1. Per-take metric scale from the IMU and from cube PnP. (a) HRL80 stacking takes, where the IMU is reliable. (b) HRA_red approach takes: the IMU scale scatters around zero.

On 7 October the operator saw the arm push the cube, so the interface now stops when the cube's apparent size in the wrist view exceeds a threshold (a proximity proxy, not verified contact): 69 of the day's 127 sessions ended this way. Execution settings mattered as much as checkpoints: an interface restart that silently reset the chunk-blending parameters made the operator report that inference quality "had dropped a lot" until they were restored. No trial outcome was logged, so there is no success rate.

VI. Lessons and Next Steps

Each scale source fails silently somewhere (IMU on slow motion, origin plane on some takes), so two independent sources are needed; a physical origin shared with the robot beats post-hoc registration; the IK TCP is a convention, not the fingertip; small single-task sets overfit at once, so early checkpoints are the ones to test; and execution settings belong to the policy. Next: logged trials with fixed settings from fixed start poses and cube positions, early checkpoints of A, B and CT7, and correcting the cube colour of the rendered images (orange in training, red on the robot).

References

[1] C. Chi *et al.*, "Universal manipulation interface," in *RSS*, 2024.
[2] R. Murai, E. Dexheimer, and A. J. Davison, "MASt3R-SLAM," in *CVPR*, 2025.
[3] J. Zheng *et al.*, "X-VLA: Soft-prompted transformer as scalable cross-embodiment VLA model," in *ICLR*, 2026.
[4] BrikHMP18, "HandUMI: Hardware," https://github.com/murobotics-ai/handumi-hw, 2026.